

Nicholas Jackson

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Education

PhD Biomedical Informatics, Vanderbilt University, Nashville, TN.
GPA: 3.93/4.0

2023 - Present

BS in Computer Science, University of Florida, Gainesville, FL.
Minor in Electrical Engineering
Summa Cum Laude - GPA: 3.90/4.0

May 2023

Relevant Coursework: Machine Learning, Mathematical Optimization, Bioinformatics, Biostatistics, Signal Processing, Databases, Generative AI, Linear Algebra

Research Experience

PhD Student, Vanderbilt Department of Biomedical Informatics

Fall 2023 - Present

Advised by Dr. Bradley Malin

- Utilized generative AI to create a multi-modal synthetic medical dataset for the AI-READI project and evaluated its privacy and utility in preparation for its public release.
- Developed and evaluated federated machine learning models to predict mortality in an international cohort of people living with HIV, assessing their sensitivity to data heterogeneity.
- Generated synthetic data to reduce disparities in the performance of medical machine learning models use to diagnose chest x-rays.

Undergraduate Researcher, Musculoskeletal Biomechanics Lab

2019 - 2023

Advised by Dr. Jennifer Nichols

- Developed an image processing pipeline to analyze fluoroscopic images and CT scans, comparing three syndesmotic alignment strategies for the purposes of surgical fixation.
- Evaluated measures of fibular length for consistency across varied image projection.

Research Intern, UF Student Science Training Program

Summer 2018

Advised by Dr. Parisa Rashidi

- Applied machine-learning to accelerometer data to identify ICU patient posture.
- Utilized unsupervised machine learning to cluster and analyze subtypes of delirium in the ICU.

Technical Skills

- **Programming Languages:** Python · MATLAB · C/C++ · Java · MySQL · R · JavaScript
- **Skills:** Statistics · Machine Learning & Artificial Intelligence · Federated Learning · Synthetic Data Generation · Algorithmic Fairness · Data Privacy Evaluation · Signal Processing

Publications - Manuscripts

Jackson NJ, Natalia ED, Yan C, et al. A Synthetic Multi-modal Dataset for Type 2 Diabetes. (*In Review*). 2026.

Jackson NJ, Yan C, Caro-Vega YN, et al. Privacy-Preserving Machine Learning for HIV: A Multi-Site Evaluation Study Across Five Countries. *SSRN*. Preprint posted online 2026. doi:10.2139/ssrn.6210360

Espinosa-Dice N, **Jackson NJ**, Yan C, Lee A, Malin BA. A Reinforcement Learning Approach to Synthetic Data Generation. *arXiv*. Preprint posted online December 24, 2025:arXiv:2512.21395. doi:10.48550/arXiv.2512.21395

Liang ZJ, Li Z, **Jackson NJ**, et al. Generating Synthetic Multi-national Longitudinal Cohorts for Clinically Grounded HIV Research. Preprint posted online November 17, 2025. doi:10.1101/2025.11.14.25340245

Borza VA, Estornell A, **Jackson NJ**, et al. The Overlooked Role of Learnability in Building Algorithmically Fair Biomedical Datasets. (*In Review*). Published online 2026.

Jackson NJ, Brown KE, Miller R, et al. Factors Influencing the Effectiveness of AI-Assisted Decision-Making in Medicine: A Scoping Review. (*In Press*) *J Am Med Inform Assoc*. 2026;00(00):1-13. doi:<https://doi.org/10.1093/jamia/ocag002>

Brown KE, Wrenn JO, **Jackson NJ**, et al. Auditor models to suppress poor artificial intelligence predictions can improve human-artificial intelligence collaborative performance. *J Am Med Inform Assoc*. Published online January 13, 2026:ocaf235. doi:10.1093/jamia/ocaf235

Jackson NJ, Andreadis K, Ancker JS. Large Language Models and Communication of Medical Probabilities. *JAMA Netw Open*. 2025;8(12):e2550449. doi:10.1001/jamanetworkopen.2025.50449

Jackson NJ, Yan C, Malin BA. Enhancement of Fairness in AI for Chest X-ray Classification. *AMIA Annu Symp Proc AMIA Symp*. 2024;2024:551-560. <https://pmc.ncbi.nlm.nih.gov/articles/PMC12099404/>

Li Z, Yan C, **Jackson NJ**, et al. Towards Statistical Factuality Guarantee for Large Vision-Language Models. In: Christodoulopoulos C, Chakraborty T, Rose C, Peng V, eds. *Proceedings of the 2025 Conference on Empirical Methods in Natural Language Processing*. Association for Computational Linguistics; 2025:11446-11467. doi:10.18653/v1/2025.emnlp-main.576

Jackson NJ. *Automated Identification of the Center-Center Axis via Deep Learning*. Undergraduate Honors Thesis. University of Florida; 2023. <https://ufdc.ufl.edu/aa00092549/00001>

Jackson NJ, Root KT, Nichols JA, Reb CW. Image Rotation Alters Apparent Fibula Length: An Evaluation of Talocrural Angle, Shenton Line, and Dime Sign. *Foot Ankle Int*. 2024;45(3):236-242. doi:10.1177/10711007231221991

Jackson NJ, Flores K, Blake A, Harley JB, Reb CW, Nichols JA. The Center-Center Image Closely Approximates Other Methods for Syndesmosis Reduction Clamp Placement. *Foot Ankle Spec*. Published online December 6, 2023:19386400231213741. doi:10.1177/19386400231213741

Conference Presentations

Jackson NJ. (2025). 'Just Give me a Number': Comparing Human and Large Language Model Interpretations of Verbal Probability Terms in Healthcare Communication. Oral Presentation presented at American Medical Informatics Association Annual Meeting. 2025 November 14; Atlanta, GA.

Jackson NJ. (2025) "Federated Learning Improves the Performance of Clinical Prediction Models for HIV Mortality". Oral Presentation presented at: International Workshop on HIV and Hepatitis Observational Databases 2025 Annual Meeting; 2025 Mar 27; Toledo, Spain.

Jackson NJ. (2024) "Enhancement of Fairness in AI for Chest X-ray Classification". Oral Presentation presented at American Medical Informatics Association Annual Meeting. 2024 November 11; San Francisco, CA.

Jackson, N. (2022) "A 3D Image-Based Comparison of the Center-Center and Transyndesmotoc Axis Methods for Patient-Specific Fixation of the Ankle Syndesmosis" Poster at Orthopedic Research Society (ORS) Annual Meeting 2022.

Jackson, N. (2021) "A Computational Comparison of Center-Center and Centroid Axes in Syndesmosis Fixation" Poster at Orthopedic Research Society (ORS) Annual Meeting. Virtual Format 2021.

Jackson, N. (2019) "Analysis of Actigraphy Data for Classifying Delirium in the ICU" Poster at Biomedical Engineering Society (BMES) Annual Meeting.

Awards and Funding

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| National Institutes of Health F31 Doctoral Fellowship , Vanderbilt University | Spring 2026- |
| ○ 7 th percentile score, awarded 3 years of funding | |
| National Library of Medicine T15 Training Grant Fellowship , Vanderbilt University | Fall 2023-Spring 2026 |
| Russell G. Hamilton Scholar , Vanderbilt University | Fall 2023- |
| Lacy-Fisher Interdisciplinary Research Grant , Vanderbilt University | 2023 – 2024 |
| Summa cum Laude , <i>University of Florida</i> | 2019 – 2023 |
| ○ >3.8 GPA + Completed Honors Thesis | |

Teaching, Service, and Leadership

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| Mentor , Vanderbilt Biomedical Informatics Summer Internship | Summer 2024 & 2025 |
| ○ Mentored undergraduate research interns for two consecutive summers, aiding in the completion of a research project. | |
| Student Leadership Committee Chair , Vanderbilt Department of Biomedical Informatics | 2024 - |
| ○ Organize department events and serve as a representative to the graduate school. | |
| Volunteer , Vanderbilt Student Volunteers for Science | Spring 2024 |
| ○ Led a series of 4 science lessons to 5 th grade students in the local Nashville Area | |
| Certificate In Engineering Leadership , University of Florida College of Engineering | Spring 2023 |
| ○ Completed 3 courses in engineering-specific leadership training. | |
| ○ Led and mentored undergraduate students during their first semester in the lab. | |
| President , University of Florida Beach Volleyball Club Team | Fall 2020 – Spring 2022 |
| ○ Implemented and maintained an allocation system for \$8,000 of resources to club members. | |
| ○ Organized a first of its kind tournament with over 150 players from 8 different universities. | |